

Ecopark Bicycle Parking

Requirement Analysis: Use-Case Analysis

Nhóm dự án hệ thống thuê xe đạp Ecopark

Tháng 5 năm 2026

Project Context and Rules

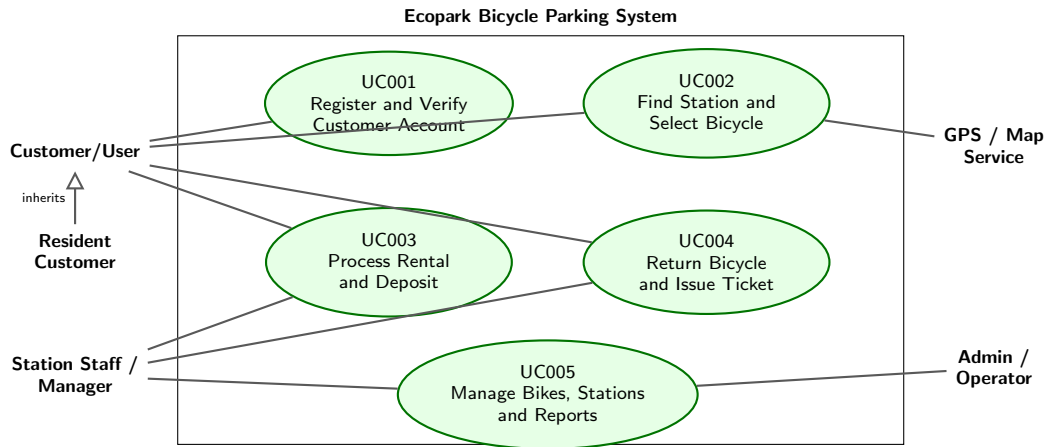
- Website quản lý thuê-trả xe đạp cho du khách và cư dân Ecopark.
- Quản lý điểm tập kết, loại xe, số lượng, ID xe duy nhất và trạng thái xe.
- Lần đầu thuê phải tạo tài khoản có email, CCCD/giấy tờ định danh, địa chỉ, số điện thoại.
- Nhận xe tại bãi với Station Staff / Manager: đặt cọc 200k + CCCD hoặc giấy tờ tùy thân.
- Giá thuê: 50k/1 giờ, 70k/2 giờ, 100k/3 giờ; trả muợn: 30k/30 phút.
- Cư dân có địa chỉ và thẻ cư dân hợp lệ được giảm 40% giá thuê.

Actor Model

Actor	Vai trò
Customer/User	Người dùng thuê xe thông thường: tạo tài khoản, tìm bãi, chọn xe, nhận/trả xe, xem vé/ lịch sử.
Resident Customer	Actor chuyên biệt kế thừa Customer/User; có thêm nhánh nộp thẻ cư dân để giảm 40%.
Station Staff / Manager	Kiểm tra giấy tờ, nhận cọc, giao xe, nhận xe trả, cập nhật trạng thái/vị trí xe tại bãi.
Admin / Operator	Quản lý toàn hệ thống: bãi xe, xe, loại xe, chính sách giá, rà soát, thống kê.
GPS / Map Service	Dịch vụ ngoài hỗ trợ tìm điểm tập kết gần nhất.

Không xem là actor: front-end, back-end, database, bảo vệ cổng.

Use-Case Overview



Relational Schema Summary

Area	Tables and relationships
Accounts	USERS → CUSTOMER_PROFILES; resident is customer_type=resident + RESIDENT_CARDS.
Fleet	STATIONS, BIKE_TYPES, BIKES, BIKE_STATUS_LOGS.
Request	CUSTOMER_PROFILES 1-n RENTAL_REQUESTS.
Rental lifecycle	RENTAL_REQUESTS 0..1-1 RENTALS; BIKES 1-n RENTALS.
Deposit and ticket	RENTALS 1-1 RENTAL_DEPOSITS; RENTALS 1-1 RENTAL_TICKETS.
Pricing	PRICING_POLICIES stores 50k/1h, 70k/2h, 100k/3h, 30k/30m and 40% discount.

- Resident Customer generalizes to Customer/User; inherited associations do not need duplicate lines.
- UC001: Submit Resident Card *extends* Create Customer Account.
- UC002: Filter Bicycle Type *extends* View Station Bicycle Inventory; Send Rental Request *includes* Select Available Bicycle.
- UC003: Confirm Bicycle Handover *includes* Verify Identity Document and Collect Deposit; Cancel Request *extends* Review Request.
- UC004: Apply Resident Discount and Apply Late Fee *extend* Calculate Rental Fee; Issue Ticket *includes* Calculate Rental Fee.
- UC005: Export Rental Reports *extends* View Rental Statistics; Update Bicycle Status *includes* Locate Bicycle.

UC001 Main Flow: Register and Verify Customer Account

Actors: Customer/User, Resident Customer, Admin / Operator for manual review.

- 1 Customer/User chooses register or log in.
- 2 System displays required account fields.
- 3 Customer/User submits account and personal information.
- 4 System validates format, duplicate email/phone and password.
- 5 System creates account and stores customer profile.
- 6 Customer/User submits CCCD/identity information.
- 7 System marks rental eligibility when identity data is valid.
- 8 Resident Customer optionally submits address and resident card.
- 9 System verifies resident evidence or sends it to review.
- 10 System displays account status, rental eligibility, discount and history.

UC001 Alternative Flows

- ❶ Missing email, CCCD, phone or address: request completion.
- ❷ Duplicate email/phone: block registration and offer login/recovery.
- ❸ Weak password: ask for stronger password.
- ❹ Invalid/blocked identity number: reject rental eligibility.
- ❺ No resident card: allow normal customer rental, no 40% discount.
- ❻ Resident card unreadable/mismatched: reject or pending discount.
- ❼ Suspicious resident profile: send to Admin / Operator review.
- ❽ Locked account: block rental request.
- ❾ Rental history unavailable: login still succeeds.
- ❿ Resident Customer inherits Customer/User and adds resident discount flow.

UC002 Main Flow: Find Station and Select Bicycle

Actors: Customer/User, GPS / Map Service.

- 1 Customer/User opens station search or rental screen.
- 2 System asks for location or manual search area.
- 3 GPS / Map Service returns location/search area.
- 4 System lists nearest stations and available bicycle counts.
- 5 Customer/User opens station detail.
- 6 System shows bike IDs, types and current statuses.
- 7 Customer/User filters type and selects an available bicycle.
- 8 System re-checks account, station and bike availability.
- 9 Customer/User confirms request and planned duration.
- 10 System creates RentalRequest and pickup instruction.

UC002 Alternative Flows

- ❶ Location permission denied: use manual station search.
- ❷ GPS / Map Service unavailable: use manual search.
- ❸ No nearby station: expand search range.
- ❹ Selected station inactive: suggest another station.
- ❺ Requested bike type unavailable: choose another type/station.
- ❻ No available bicycle: block request.
- ❼ Selected bicycle becomes rented/broken: refresh and reselect.
- ❽ Account lacks identity data: redirect to UC001.
- ❾ User cancels before confirmation: no RentalRequest.
- ❿ Request expires before pickup: cancel and release bicycle.

UC003 Main Flow: Process Bicycle Rental and Deposit

Actors: Customer/User, Station Staff / Manager.

- 1 Customer/User shows RentalRequest code at pickup station.
- 2 Staff searches pending request.
- 3 System displays customer, bike, station and planned duration.
- 4 Staff checks CCCD/ID or accepted identity document.
- 5 Staff confirms document matches customer account.
- 6 Staff collects 200k deposit and records held document.
- 7 System re-checks station and bicycle status.
- 8 Staff hands over the correct bicycle.
- 9 System creates Rental, deposit record and start time.
- 10 System marks bicycle rented and shows handover receipt.

UC003 Alternative Flows

- ❶ RentalRequest not found/expired: ask user to create new request.
- ❷ Request belongs to another station: block current processing.
- ❸ Customer forgot identity document: no handover.
- ❹ Document mismatch: reject and mark for review.
- ❺ Deposit below 200k: do not create rental.
- ❻ Customer refuses to leave document: do not hand over bicycle.
- ❼ Bicycle becomes broken/unavailable: block handover.
- ❽ Bicycle not at expected station: flag data mismatch.
- ❾ Customer wants another bike: return to UC002.
- ❿ Rental/deposit save fails: retry; do not mark bike rented.
- ⓫ Receipt display/print fails: rental remains stored.

UC004 Main Flow: Return Bicycle and Issue Ticket

Actors: Customer/User, Station Staff / Manager.

- 1 Customer/User brings bicycle to any Ecopark station.
- 2 Staff searches rental by rental ID, customer or bike ID.
- 3 System displays rental, pickup station, start time and deposit.
- 4 Staff inspects returned bicycle condition.
- 5 Staff confirms return station and return time.
- 6 System updates return station and actual duration.
- 7 System calculates base fee: 50k/1h, 70k/2h, 100k/3h.
- 8 System applies 40% resident discount when eligible.
- 9 System adds 30k/30min late fee when returned late.
- 10 System issues ticket, stores history and updates bike status/location.

UC004 Alternative Flows

- ➊ Return station differs from pickup: allow and update location.
- ➋ No in-progress rental found: manual operation review.
- ➌ Deposit mismatch: flag reconciliation.
- ➍ Bicycle damaged/needs repair: mark broken, not available.
- ➎ Staff entered wrong station/time: allow correction before ticket.
- ➏ Actual time below planned slot: apply configured price policy.
- ➐ Resident card not verified: no 40% discount.
- ➑ Late time is partial 30 minutes: round by late-fee policy.
- ➒ Customer disputes late fee/time: issue ticket and flag review.
- ➓ Ticket cannot print/send: ticket remains stored.
- ➔ Status/location update fails: flag bike data for operations.

UC005 Main Flow: Manage Bikes, Stations and Rental Reports

Actors: Admin / Operator, Station Staff / Manager.

- ➊ Actor opens operation management.
- ➋ Actor chooses station, bicycle, status, locate bike or reports.
- ➌ Station action: create/update name, address, coordinates, capacity, status.
- ➍ Bicycle action: create/update unique ID, type, current station and status.
- ➎ Status action: select bicycle and new operational state.
- ➏ System logs status/location change with actor and reason.
- ➐ Locate bike: search by bike ID or filters.
- ➑ Statistics: choose period, station, bike type or bike ID.
- ➒ System aggregates rentals, revenue, late fees, returns and statuses.
- ➓ System displays dashboard or exports report.

UC005 Alternative Flows

- 1 Staff lacks global permission: limit to assigned station.
- 2 New station missing fields: reject save.
- 3 Capacity below current bike count: reject update.
- 4 Bike ID already exists: reject create/update.
- 5 Rented bike manual station change: block until rental ends.
- 6 Bike marked broken: hide from available list.
- 7 Broken bike reopened: require inspection/repair note.
- 8 Bike search has no result: show empty result.
- 9 Report filters have no data: show empty report.
- 10 Rentals pending reconciliation: mark statistics provisional.
- 11 Report export fails: keep filters and allow retry.

Presentation Checklist

- Start with overview diagram, actor hierarchy, and why 5 UCs were selected.
- For each UC, state actor, goal, observable result, main flow and alternative flows.
- Explain 'include'/'extend' only where the relation changes the interpretation of the diagram.
- Emphasize common mistakes avoided: no front-end/back-end/database actors; no action-only UCs.
- Use A4 PDF as SRS-style detailed backup for inputs, outputs and acceptance rules.